

GWC Academy

JAVA PROGRAMMING & OBJECT-ORIENTED PROGRAMMING

Live Instructor-Led Course Curriculum | Beginner to Intermediate | Practical & Project-Based

Course Duration	60 Hours
Recommended Duration	10–12 Weeks
Mode	Live Online / Classroom
Level	Beginner to Intermediate
Prerequisites	Basic computer knowledge; no prior programming experience required
Tools	JDK, IntelliJ IDEA or Eclipse, Git/GitHub, MySQL, Maven
Outcome	Build strong Core Java, OOP, Collections, File Handling, JDBC and multithreading foundations

Course Overview

This curriculum is designed to take learners from programming fundamentals to practical Java development. The course focuses on modern, industry-relevant Core Java concepts, strong OOP fundamentals, debugging, exception handling, collections, file handling, multithreading, database connectivity and real-world projects.

Detailed Module-Wise Curriculum

Module 1: Introduction to Programming & Java (2 Hours)

- Introduction to programming and problem-solving
- What is Java and where it is used
- History and evolution of Java
- JDK, JRE and JVM
- How Java programs compile and run
- Java features: platform independence, robustness, security and portability
- Java versions and LTS concept
- Installing JDK and configuring environment variables

Module 2: Java Development Environment (2 Hours)

- Introduction to IDEs
- IntelliJ IDEA / Eclipse setup
- Creating and organizing Java projects
- Source files, packages and class structure
- Compile and run Java applications
- Understanding syntax, semantic and runtime errors
- Debugging basics and breakpoints

Module 3: Java Fundamentals (5 Hours)

- Hello World program and program structure
- Keywords, identifiers and naming conventions
- Variables and constants
- Primitive and reference data types
- Type conversion and type casting
- Operators: arithmetic, relational, logical, assignment, unary and ternary
- Input using Scanner and BufferedReader
- Command-line arguments

Module 4: Control Flow & Problem Solving (5 Hours)

- Conditional statements: if, if-else and nested conditions
- switch statement and switch expressions
- Loops: for, while and do-while
- break, continue and return
- Enhanced for loop
- Pattern and logic-building exercises
- Basic algorithmic problem-solving practice

Module 5: Arrays & Strings (5 Hours)

- One-dimensional arrays
- Multi-dimensional arrays
- Array traversal, searching and basic operations
- Common array-based programming problems
- String creation and immutability
- String methods and comparisons
- StringBuilder and StringBuffer
- Basic text-processing problems

Module 6: Methods, Scope & Recursion (4 Hours)

- Method declaration and invocation
- Parameters and return types
- Method overloading
- Pass-by-value behavior in Java
- Variable scope and lifetime
- static methods and members
- Introduction to recursion
- Practice programs using methods and recursion

Module 7: Object-Oriented Programming Fundamentals (6 Hours)

- Procedural programming vs object-oriented programming
- Classes and objects
- Fields, methods and object state
- Constructors and constructor chaining
- this and super keywords
- Access modifiers
- Encapsulation and data hiding
- Real-world OOP modeling exercises

Module 8: Advanced OOP Concepts (6 Hours)

- Inheritance and types of inheritance
- Method overriding
- Polymorphism: compile-time and runtime
- Abstraction
- Abstract classes and abstract methods
- Interfaces and interface implementation
- Multiple interface implementation
- Association, aggregation and composition
- final keyword

Module 9: Packages, Inner Classes & Java Core APIs (3 Hours)

- Creating and using packages
- Import statements and classpath basics
- java.lang package overview
- Object class: toString(), equals() and hashCode()
- Wrapper classes and autoboxing/unboxing
- Math, Random and utility classes
- Introduction to inner and nested classes

Module 10: Exception Handling & Debugging (4 Hours)

- Errors vs exceptions
- Checked and unchecked exceptions
- try, catch and finally
- Multiple catch blocks
- throw and throws
- Custom/user-defined exceptions
- try-with-resources
- Exception-handling best practices
- Debugging common runtime issues

Module 11: File Handling & Input/Output (4 Hours)

- Introduction to Java I/O
- File and Path concepts
- Byte streams vs character streams
- FileInputStream and FileOutputStream
- FileReader and FileWriter
- BufferedReader and BufferedWriter
- Creating, reading, writing, updating and deleting files
- Introduction to serialization

Module 12: Collections & Generics (6 Hours)

- Introduction to the Collection Framework
- Generics and type safety
- List: ArrayList and LinkedList
- Set: HashSet, LinkedHashSet and TreeSet
- Map: HashMap, LinkedHashMap and TreeMap
- Queue, PriorityQueue and Deque
- Stack and legacy collections overview
- Iterator and enhanced for loop
- Comparable and Comparator
- Collections utility methods

Module 13: Multithreading & Concurrency Basics (4 Hours)

- Process vs thread
- Thread life cycle
- Creating threads using Thread and Runnable
- Thread methods and states
- Synchronization
- Race conditions and thread interference
- Deadlock concept and prevention basics
- Thread-safe programming principles
- Introduction to ExecutorService

Module 14: JDBC & MySQL Integration (5 Hours)

- Introduction to databases and JDBC architecture
- Installing and connecting MySQL
- JDBC driver and connection setup
- Connection, Statement and PreparedStatement
- CRUD operations
- ResultSet processing
- Transactions and batch operations
- Building a database-driven Java application
- Basic login/user-management flow

Module 15: Modern Java Essentials (3 Hours)

- Introduction to Java 8+ features
- Lambda expressions
- Functional interfaces
- Method references
- Optional basics
- Stream API basics
- Date and Time API
- Writing cleaner Java code

Module 16: Build, Version Control & Capstone Project (6 Hours)

- Introduction to Maven and dependency management
- Project structure and basic build lifecycle
- Introduction to Git and GitHub
- Repository creation, commit and push workflow
- Code organization and clean coding basics
- Capstone project development
- Testing, debugging and project presentation
- Interview-oriented revision and coding practice

Practical Learning & Assessments

Hands-On Practice: Every major module should include coding exercises, logic problems and instructor-guided implementation.

Module Assignments: Short assignments covering variables, control flow, arrays, OOP, exceptions, collections, files, multithreading and JDBC.

Mini Projects: Suggested projects: Student Management System, Bank Account Simulator, Library Management System or Employee Management System.

Capstone Project: A database-driven Java application using OOP, Collections, Exception Handling, File Handling where applicable, JDBC and MySQL.

Interview Preparation: Java fundamentals, OOP concepts, collections, exceptions, multithreading and JDBC interview questions with coding practice.

Recommended Pricing

Plan	Recommended Fee	Suitable For
Launch / Early Bird	■7,999	First cohorts and introductory offer
Regular Live Batch	■9,999	Recommended main website selling price
Premium Mentorship	■14,999	Includes additional project review and priority doubt support

Recommended Website Strategy: Display ■9,999 as the regular course fee and use ■7,999 as a limited launch/early-bird offer for initial batches.

What Is Not Included in This Course

Advanced enterprise Java topics such as Spring Boot, Hibernate/JPA, REST API development, microservices, Servlet/JSP and full-stack Java development are better structured as separate follow-up courses. Legacy technologies such as Applets are intentionally not part of the core curriculum.

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Learn. Build. Grow.